S2 Supplementary Data

Frozen primary reliability and longitudinal estimates

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S2.1 Primary test–retest results

In the cohort-anchored 13-child session 1–2 comparison, reachability condition number had ICC(A,1)=0.946 (bootstrap 95% CI −0.132 to 0.990), reachability log-determinant had 0.779 (0.216 to 0.942), and directional entropy had 0.535 (-0.160 to 0.826). MNJ scalar estimates were low or negative. The FDR column below adjusts the nine paired Wilcoxon session-difference p-values; it does not test ICC. No paired session difference survived this sensitivity analysis (all adjusted p >= 0.945). These small-sample intervals remain broad and should not be treated as precise reliability classification.

Endpoint	ICC	95% CI	r	SEM	MDC95	Mean Δ	Raw p	FDR p
frobenius_norm	0.064	−0.360, 0.421	0.062	0.018	0.050	0.0032	0.542	0.945
rotation_norm	0.119	−0.380, 0.449	0.115	0.013	0.036	0.0031	0.685	0.945
trace	−0.074	−0.515, 0.212	−0.068	0.013	0.035	0.0002	0.946	0.946
spectral_radius	−0.290	−0.762, 0.199	−0.270	0.008	0.024	0.0017	0.735	0.945
tube_log_det	0.779	0.216, 0.942	0.769	0.253	0.701	−0.0502	0.414	0.945
tube_d_eff	0.442	−0.158, 0.771	0.471	0.150	0.416	0.0614	0.216	0.945
tube_kappa	0.946	−0.132, 0.990	0.941	1.644	4.558	0.0569	0.735	0.945
tube_dir_entropy	0.535	−0.160, 0.826	0.562	0.062	0.172	0.0247	0.340	0.945
tube_rotation	0.217	−0.545, 0.627	0.204	0.084	0.233	−0.0056	0.946	0.946

Table 1: Primary absolute-agreement reliability results. Each line uses 13 paired children. Mean Δ is session 2 minus session 1; SEM and MDC95 retain endpoint units. FDR is BH adjustment within the nine paired session-difference tests.

S2.2 Cohort-anchored 4-to-8-month dynamical contrasts

Eleven complete children contributed the primary cohort-anchored four-month mean to eight-month panel. No primary endpoint survived the nine-test BH-FDR sensitivity correction; the smallest raw p-values were 0.102 for lower reachability effective dimension and lower directional entropy. The table reports all nine endpoints, not only the post-hoc display subset.

Endpoint	Contrast	Mean difference	d_z	Raw p / FDR p
frobenius_norm	4-month mean to 8 months	0.0090	0.34	0.240 / 0.309
rotation_norm	4-month mean to 8 months	0.0061	0.35	0.206 / 0.309
trace	4-month mean to 8 months	0.0063	0.37	0.413 / 0.413
spectral_radius	4-month mean to 8 months	0.00493	0.47	0.175 / 0.309
tube_log_det	4-month mean to 8 months	0.198	0.36	0.175 / 0.309
tube_d_eff	4-month mean to 8 months	−0.096	−0.48	0.102 / 0.309
tube_kappa	4-month mean to 8 months	0.373	0.04	0.147 / 0.309
tube_dir_entropy	4-month mean to 8 months	−0.0425	−0.39	0.102 / 0.309
tube_rotation	4-month mean to 8 months	0.0333	0.37	0.278 / 0.313

Table 2: Complete nine-endpoint cohort-anchored 4-month mean to 8-month family. Raw p-values are exact two-sided Wilcoxon signed-rank values; FDR is BH within these nine tests. Bootstrap intervals for mean differences are in the machine-readable CSV.

S2.3 Cohort-anchored 4-to-12-month dynamical contrasts

The same 11 complete children contributed the four-month mean to twelve-month panel. None of the nine 3D dynamical endpoints had raw p below 0.365 and none survived the nine-test BH-FDR sensitivity correction.

Endpoint	Contrast	Mean difference	d_z	Raw p / FDR p
frobenius_norm	4-month mean to 12 months	0.0209	0.45	0.638 / 0.717
rotation_norm	4-month mean to 12 months	0.0143	0.45	0.465 / 0.717
trace	4-month mean to 12 months	0.0075	0.29	0.465 / 0.717
spectral_radius	4-month mean to 12 months	0.0057	0.37	0.638 / 0.717
tube_log_det	4-month mean to 12 months	0.0284	0.04	0.831 / 0.831
tube_d_eff	4-month mean to 12 months	−0.0720	−0.35	0.577 / 0.717
tube_kappa	4-month mean to 12 months	2.468	0.37	0.520 / 0.717
tube_dir_entropy	4-month mean to 12 months	−0.0365	−0.34	0.365 / 0.717
tube_rotation	4-month mean to 12 months	0.0233	0.20	0.520 / 0.717

Table 3: Complete nine-endpoint cohort-anchored 4-month mean to 12-month family. FDR is BH within these nine tests.

S2.4 Coordinate-position and comparator reliability

Coordinate positions were not used as a substitute for dynamical validation. Their session 1–2 ICC(A,1) estimates were 0.575 for mobility (bootstrap 95% CI 0.038 to 0.895), 0.458 for deviation (0.072 to 0.706), and 0.700 for entropy-related coordinate (0.019 to 0.894); these intervals provide context for the 12-month mobility contrast. In that separate three-coordinate family, mobility decreased in all 11 children (raw p=0.00098; BH p=0.0029).

Coordinate	ICC	Bootstrap 95% CI	r
Mobility	0.575	0.038, 0.895	0.629
Deviation	0.458	0.072, 0.706	0.603
Entropy-related	0.700	0.019, 0.894	0.686

Table 4: Descriptive session 1–2 reliability of the three cohort-anchored coordinate positions; n=13.

The conventional EEG comparator endpoints exported by the same pipeline were also treated descriptively. Session 1–2 ICC estimates were approximately 0.41 for alpha/theta ratio, 0.76 for beta/alpha ratio, 0.45 for relative delta, and 0.79 for permutation entropy; complete values, intervals, SEM, and paired tests are in `reliability_conventional.csv`.

Comparator	ICC	Bootstrap 95% CI	r	FDR p
alpha/theta ratio	0.405	0.058, 0.687	0.498	0.751
beta/alpha ratio	0.763	0.545, 0.879	0.799	0.751
relative delta	0.455	−0.008, 0.651	0.453	0.787
permutation entropy	0.793	−0.212, 0.941	0.803	0.787

Table 5: Descriptive reliability of conventional EEG comparators exported by the same pipeline; FDR is within this four-endpoint family.

S2.5 Nine-dimensional sensitivity and event outputs

The 9D Jacobian analysis tests ten exported summaries rather than the 3D reachability family. Its complete cases are n=10 for 4-to-8 months and n=9 for 4-to-12 months; the lowest raw p-values were 0.193 and 0.039, respectively, with separate-family FDR p-values of 1.000 and 0.391. The 4-to-8 loss is driven by `sub-13/ses-2` (155 invalid windows; zero retained), and the 4-to-12 loss additionally includes `sub-10/ses-4` (140 invalid windows; zero retained). No 9D value was imputed.

The event analysis pools available ages after averaging event-minus-`pre_near` contrasts within child. All 12 event-by-axis tests used 13 children, but the number of contributing sessions and minimum stage-label fraction are reported in `event_exploratory_tests.csv`. The water-deviation contrast had raw permutation p=0.0055 and BH p=0.066; it remains exploratory and is not evidence for a developmental or mechanistic effect.